

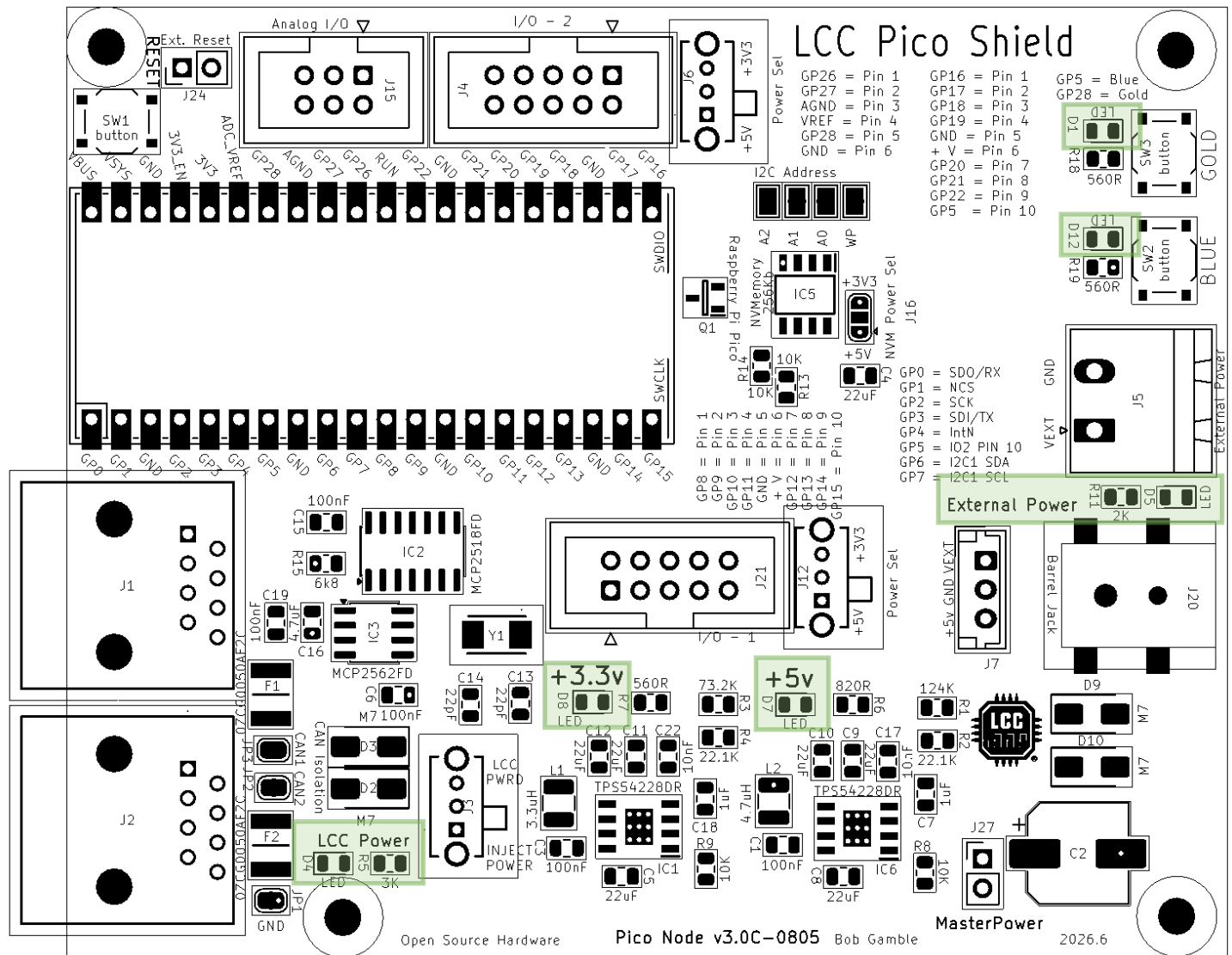
LCC Pico Node User Guide

Version 3.0

This version utilizes the MCP2815 and MCP2562 discrete hardware components (chips) for the LCC interface.

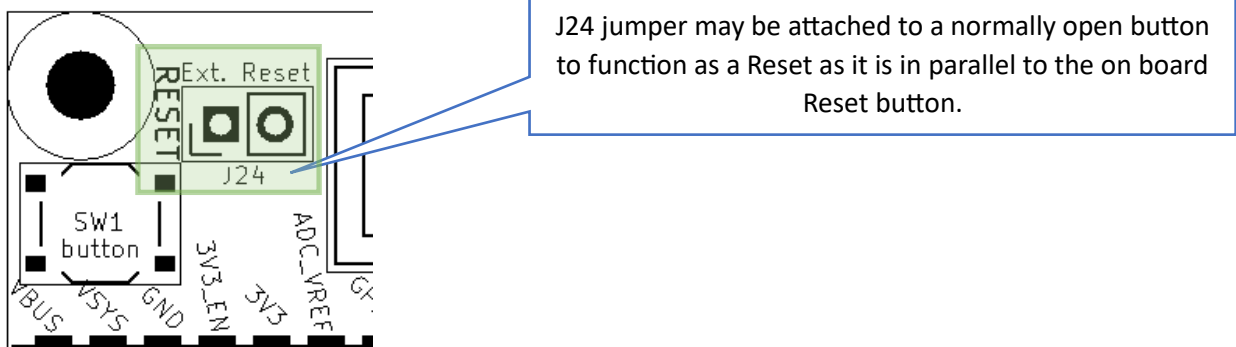
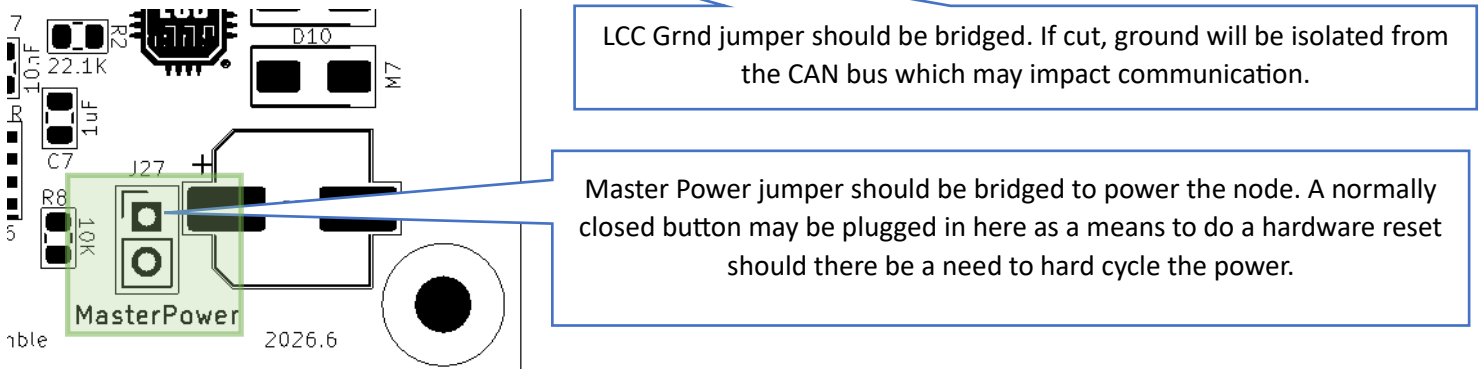
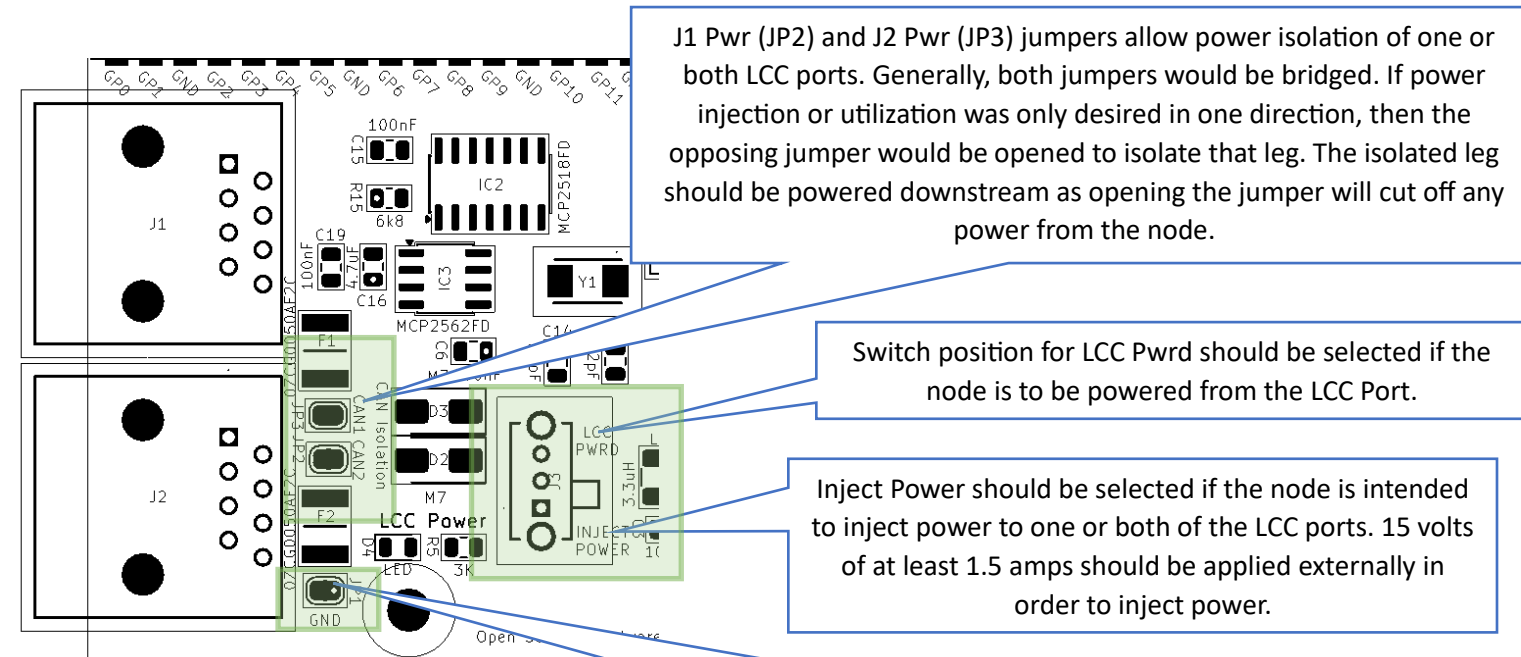
The intent of this node is to control 16 pins on two 10 pin IDC connectors for a model railroad layout. It exposes 16 GP pins through two 10P IDC connectors, plus an additional 6-pin analog header (I/O-3), for general-purpose I/O. While it will work with either the Pico or Pico2, the latter is recommended due to enhanced capabilities.

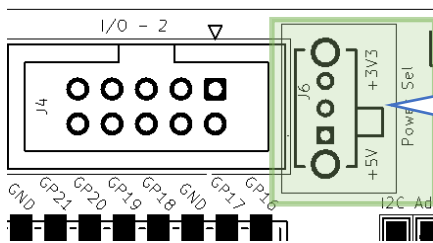
There are no resistors on the Reset, Blue, and Gold button lines other than for the Blue and Gold LEDs.



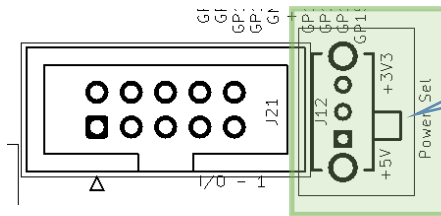
The node may be powered from the barrel/terminal jack or the LCC port. The barrel jack power should be at least eight (8) volts to power the microcontroller or fifteen (15) volts [or your preferred LCC voltage] if using the node to inject power to LCC. Amperage should be sufficient to drive any external devices plus about half an amp for the node plus another 1.5 amps if injecting power to LCC. There are four power LEDs. The 3.3- and 5-volt LEDs should be lit to indicate that the node is properly powered.

Node power configuration is accomplished with five jumpers located behind the LCC ports.





The power applied to the middle pins of the IDC connector may be selected by the power selection switch. Pin assignments are noted on the PCB.



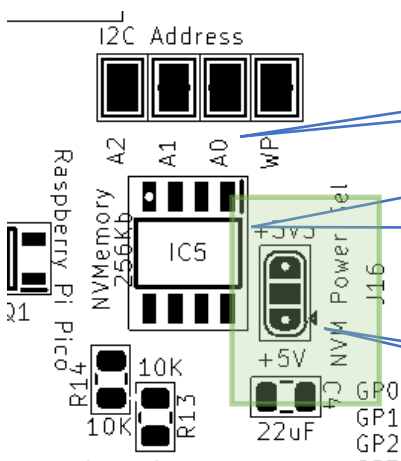
The power applied to the middle pins of the IDC connector may be selected by the power selection switch and the solder jumper. Pin assignments are noted on the PCB.

NVM Storage

I2C nonvolatile memory is used to reduce strain on the Pico flash. External NVM is soldered down to the IC5 footprint. The I2C Address jumpers may be used to select non-default addresses or implement write protection. Otherwise, these would be functions of the module plugged into the I2C header.

Parts compatible with the following should work for soldering to the footprint on the board.

- BR24G256F-3AGTE2 I2C BUS(2-Wre) 256K or other sizes with equivalent pinouts
- MB85RC256VPF-G-BCERE1 256kbit FRAM with I2C



I2C address may be selected by the left three jumpers.

I2C nonvolatile memory may be populated if desired to reduce strain on the Pico flash. Parts compatible with these should work.

BR24G256F-3AGTE2 I2C BUS(2-Wre) 256K

MB85RC256VPF-G-BCERE1 256kbit FRAM with I2C

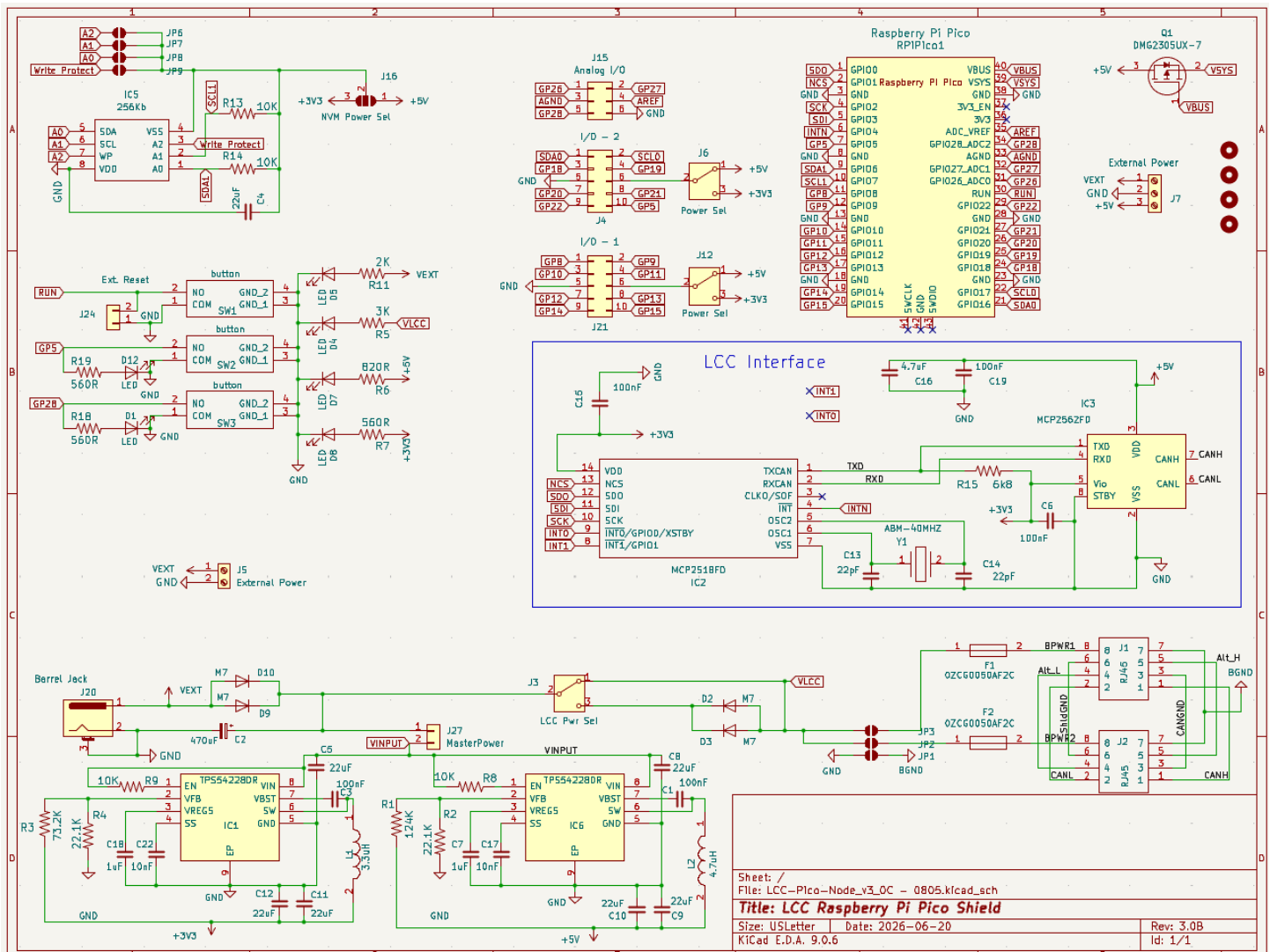
3.3V is the preferred selection for this solder jumper. The selection will apply to the soldered NVM module.

Board Pin Assignments

PIN	GP	ALT 1	ALT 2	V3.0 NODE
1	gp0	SPI0 RX	I2C0 SDA	ACAN_RX_PIN
2	gp1	SPI0 CSN	I2C0 SCL	ACAN_CS_PIN
3	GND			
4	gp2	SPI0 SCK	I2C1 SDA	ACAN_CSK_PIN
5	gp3	SPI0 TX	I2C1 SCL	ACAN_TX_PIN
6	gp4	SPI0 RX	I2C0 SDA	ACAN_INT_PIN
7	gp5	SPI0 CSn	I2C0 SCL	I/O-2: Pin 10 & Blue Button
8	GND			
9	gp6	SPI0 SCK	I2C1 SDA	I2C1 SDA
10	gp7	SPI0 TX	I2C1 SCL	I2C1 SCL
11	gp8	SPI1 RX	I2C0 SDA	I/O-1: Pin 1
12	gp9	SPI1 CSn	I2C0 SCL	I/O-1: Pin 2
13	GND			
14	gp10	SPI1 SCK	I2C1 SDA	I/O-1: Pin 3
15	gp11	SPI1 TX	I2C1 SCL	I/O-1: Pin 4
16	gp12	SPI1 RX	I2C0 SDA	I/O-1: Pin 7
17	gp13	SPI1 CSn	I2C0 SCL	I/O-1: Pin 8
18	GND			
19	gp14	SPI1 SCK	I2C1 SDA	I/O-1: Pin 9
20	gp15	SPI1 TX	I2C1 SCL	I/O-1: Pin 10
21	gp16	SPI0 RX	I2C0 SDA	I/O-2: Pin 1
22	gp17	SPI0 CSn	I2C0 SCL	I/O-2: Pin 2
23	GND			
24	gp18	SPI0 SCK	I2C1 SDA	I/O-2: Pin 3
25	gp19	SPI0 TX	I2C1 SCL	I/O-2: Pin 4
26	gp20		I2C0 SDA	I/O-2: Pin 7
27	gp21		I2C0 SCL	I/O-2: Pin 8
28	GND			
29	gp22			I/O-2: Pin 9
30	RUN			Reset Button / header

31	gp26	ADC0	I2C1 SDA	I/O-3: Pin 1
32	gp27	ADC1	I2C1 SCL	I/O-3: Pin 2
33	GND	A GND		I/O-3: Pin 3 (AGND)
34	gp28	ADC2		I/O-3: Pin 5 & Gold Button
35		ADC_VREF		I/O-3: Pin 4 (VREF)
36	3v3_OUT			
37	3V3_EN			
38	GND			
39	VSYS			
40	VBUS			
	GND		I/O-1: Pin 5	
	Vselect1		I/O-1: Pin 6	
	GND		I/O-2: Pin 5	
	Vselect2		I/O-2: Pin 6	

Note: I/O-2 Pin 10 (gp5) and I/O-3 Pin 5 (gp28) double as the Blue and Gold buttons. A breakout that uses either of those pins makes the corresponding button unavailable.



Bill of Materials

Qty	Value	Reference	Mfr. #	Description
5	0.1uF, 100nF	C1,C3,C6,C15,C19	C1206C104K3RAC7411	Multilayer Ceramic Capacitors MLCC - SMD/SMT 25V 0.1uF X7R 1206 10%
1	470uF	C2	REF1016471M035K	Aluminum Electrolytic Capacitor
7	22uF	C4,C5,C8,C9,C10,C11,C12	CL31A226MOCLNNC	Multilayer Ceramic Capacitors MLCC - SMD/SMT 22uF+/-20% 16V X5R 3 1206
2	1uF	C7, C18	CL31B105KAHNFNE	Multilayer Ceramic Capacitors MLCC - SMD/SMT 1uF+/-10% 25V X7R 32 1206
2	22pF	C13,C14	C1206C220J3HACTU	Multilayer Ceramic Capacitors MLCC - SMD/SMT 25V 22pF X8R 1206 5%
1	4.7uF	C16	CL31B475KOHNFNE	Multilayer Ceramic Capacitors MLCC - SMD/SMT 4.7uF+/-10% 16V X7R 1206
2	10nF	C17, C22		
6	LED	D1,D4,D5,D7,D8,D12	156120VS75000	Single Color LEDs WL-SMRW SMD RvMt Mon Wtrclr Rect 1206BtGn
4	M7	D2,D3,D9,D10	M7	Rectifiers Diode, SMA, 1000V, 1A
2	OZCG0050AF2C	F1,F2	OZCG0050AF2C	Resettable Fuses - PPTC
1	TPS54228DDAR	IC1, IC6	TPS54228DDAR	4.5V to 18V Input, 2-A Synchronous Step-Down Converter with Eco-mode™
1	MCP2518FDT-E_SL	IC2	MCP2518FDT-E/SL	CAN Interface IC Stand-alone Low Power CAN FD Controller w/SPI Interface Grade1
1	MCP2562FDT-E/SN	IC3	MCP2562FDT-E/SN	CAN Interface IC CAN Flexible Data Rate Transceiver
1	TPS3840DxNBR	IC4		
1	256Kb	IC5	BR24G256F-5E2	EEPROM I2C BUS(2-Wre) 256K SOP8 EEPROM
2	RJ45	J1,J2	MTJ-880X1	Modular Connectors / Ethernet Connectors 8P8C TYPE 0 GLD FLSH BLK
1	3.3uH	L1	1277AS-H-3R3M=P2	Power Inductors - SMD Ind,3.2x2.5x1mm,3.3uH+/-20%,4A, shd AEC-Q200
1	4.7uH	L2	1276AS-H-4R7M=P2	Power Inductors - SMD Ind,3.2x2.5x1mm,4.7uH+/-20%,3.1A, shd AEC-Q200
1	DMG2305UX-7	Q1	DMG2305UX-7	MOSFETs P-Ch ENH FET -20V 52mOh -5A
1	124K	R1	ERA-8AEB1243V	Thin Film Resistors - SMD 1206 124Kohm 25ppm 0.1% AEC-Q200
2	22.1K	R2,R4	ERA-8AEB2212V	Thin Film Resistors - SMD 1206 22.1Kohm 25ppm 0.1% AEC-Q200
1	73.2K	R3	ERA-8AEB7322V	Thin Film Resistors - SMD 1206 73.2Kohm 25ppm 0.1% AEC-Q200
1	3K	R5		
1	820R	R6	RT1206FRE07820RL	Thin Film Resistors - SMD 1/4W 820 ohm 1% 50ppm
3	560R	R7,R18,R19	RT1206FRE07560RL	Thin Film Resistors - SMD 560 OHM 1% 50PPM 1/4W
4	10K	R8,R9,R13,R14		
1	2K	R11	RT1206FRE072KL	Thin Film Resistors - SMD 1/4W 2K ohm 1% 50ppm
1	6k8	R15	RT1206DRD076K8L	Thin Film Resistors - SMD 6.8 kOhms 250 mW 0.5 % 1206
1	Raspberry Pi Pico	RPiPico1		
3	button	SW1, SW2, SW3		Tactile Switch
1	ABM-40MHZ	Y1	ABM3-40.000MHZ-B2-T	Crystals Xtal 5032 2-SMD 40MHz Tol +/-20ppm Stab +/-50ppm -20C - 70C 18pF 25 Ohms

Connections:

J3	Select power from LCC or Inject Power to LCC
J4, J21	10 pin IDC connector I/O
J6,J12	3.3/5 Volt Power Selection
J16	FRAM Power Select
J7	power out
J5	Power terminal block
J20	Barrel Jack power input

J24	External Reset header
J27	MasterPower bridge header
JP2	J2 LCC Power enable
JP3	J1 LCC Power enable
JP1	LCC Ground enable
JP6-9	NVM Address & WP

Software

Example Node Applications

- https://github.com/bocabob/LCC_RPiPico_Turntable
- https://github.com/bocabob/LCC_RPiPico_Roundhouse
- https://github.com/bocabob/LCC_RPiPico_PixelLights
- https://github.com/bocabob/LCC_RPiPico_Clock_Lights

Architectural Standard

Cross-project firmware conventions (board versioning, breakout pin assignments, configuration memory, naming) for all node applications above are documented in: https://github.com/bocabob/LCC_RPiPico_Common

Board Support

Use the RP2040 board library by Earle Philhower: <https://arduino-pico.readthedocs.io/en/latest/install.html>

LCC Support

- https://github.com/openlcb/OpenLCB_Single_Thread
- <https://github.com/JimKueneman/OpenLcbCLib>

CAN Hardware

- ACAN2517 Arduino library

External Storage

- I2C_EEPROM Arduino library by Rob Tillaart
- FRAM_I2C Arduino library by Rob Tillaart

NeoPixel

- NeoPixelBus by Makuna (preferred)
- NeoPixelConnect by Alan Yorinks

Other Preferred Libraries

Display

- TFT_eSPI_RA8876.h — display library (modified): https://github.com/bocabob/TFT_eSPI_RA8876
- RA8876_RP2040.h — display library (modified): https://github.com/bocabob/RA8876_RP2040
- my_bb_captouch.h — capacitive touch driver (modified): https://github.com/bocabob/my_bb_captouch

Servos

- ServoEasing.hpp — for slow servo action, including bounces
- PCA9685_servo_driver.h
- PCA9685_servo.h

Stepper Motor

- AccelStepper.h by Mike McCauley (modified)